HARARE INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

B.TECH (HONS) DEGREE IN COMPUTER SCIENCE

B.TECH (HONS) DEGREE IN SOFTWARE ENGINEERING

ICS 2103/ ISE 2103: DATA COMMUNICATION AND NETWORKS

TIME: 3 HOURS **DATE: NOV/DEC 2023**

TOTAL MARKS: 100

INSTRUCTIONS TO CANDIDATES:

- *This question paper contains a total of **FIVE** questions.*
- *Each question carries **25** marks*
- *Answer **any FOUR** questions*
- *Start a new question on a fresh page*

ADDITIONAL MATERIALS:

- *None*

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Question 1

a. The figure below depicts the format of the IP header.

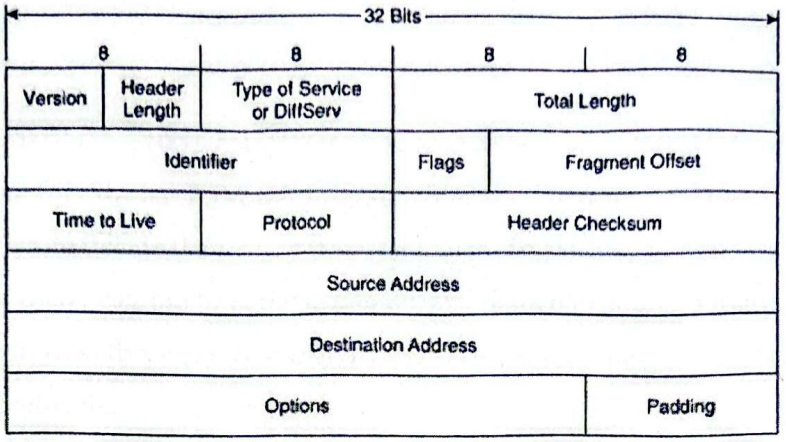


Figure 1: IP Header

State the purpose of the following IPv4 header fields;

i. Version. **[1]**

ii. Differentiated services. **[1]**

iii. Time-to-live. **[1]**

iv. Protocol. **[1]**

v. Source IP address. **[1]**

vi. Destination IP address. **[1]**

b. Indicate **five (5)** advantages of IPv6 over IPv4. **[5]**

c. Compare and contrast the key differences between the OSI Layer model and the TCP/IP Reference model. **[4]**

d. Considering IPv4 classless addressing, an organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subnets of addresses to use in its three subnets as shown below:

- One subnet of 120 addresses.
- One subnet of 60 addresses.
- One subnet of 10 addresses.

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Show the network IP address and the subnet mask for each of the subnet using Variable Length Subnet Masking (VLSM). Show all working. [6]

e. List any **four (4)** types of transmission media that you are familiar with. [4]

Question 2

a. Natalie intends to setup a network with several routers purchased by the organization she is working for. She has however, forgotten how routing tables work and needs this information to decide the routing protocol to use. To aid her, describe **three (3)** different sources a router can use to put/maintain information contained in a routing table. [3]

b. Tabulate at least **three (3)** main differences between link-state and distance-vector routing protocols. [6]

c. Copy and complete Table 1 below;

Table 1: Routing.

Routing Protocol	**Distance-vector or Link-state?**	**Metric used**
RIPv1		
EIGRP		
OSPF		

[6]

d. Describe the workings of the EIGRP routing protocol, clearly highlighting how it selects the best path for transmission of packets. [5]

e. Discuss the quality-of-service requirements of a Voice-over-IP (VoIP) application and how they differ from those of a video-based application. [5]

Question 3

a. Using a diagram, explain how data is transmitted along a fiber optic cable. [5]

b. Identify **three (3)** physical characteristics of fiber optic cables that make them more suitable for high-speed digital data transmission over copper cables. [6]

c. A host was given the IP address 198.168.4.121/27. Consider this address and indicate:

i. The network address to which the host belongs. [1]

ii. The network broadcast address to which the host belongs. [1]

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iii. The total number of hosts available in the network. [1]

iv. Determine if the following IP address be assigned as the default gateway of the host: 192.168.4.93. Give an explanation. [1]

d. The Internet is often described as being a "best effort network". Explain what is meant by the term "best effort network". [4]

e. List three QoS parameters that are often measured to characterize the behavior of a network or network connection. [6]

Question 4

a. Using a table, differentiate CSMA/CD from CSMA/CA. [5]

b. Copy and complete the Table 2 below;

Table 2: Errors

Type of error	Description	Detection Method
Single bit		
Burst error		

[4]

c. Highlight the major differences between DES and AES encryption methods. [4]

d. Define the following terms as applied in network security;

i. Key distribution center. [3]

ii. Certification authority. [3]

e. The transmission mode defines the direction of signal flow between two connected devices. State the **three (3)** transmission modes, giving examples for each mode. [6]

Question 5

a. The data link layer in the IEEE standard is divided into two (2) sub layers. Discuss the functions performed by:

i. LLC. [3]

ii. MAC. [3]

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b. Switching, as applied to networking and IT, is the practice of directing a signal or data element toward a particular hardware destination.

i. Explain the main difference between a circuit-switched network and a packet switched network. **[4]**

ii. Give **two (2)** examples of packet-switched protocols and **two (2)** examples of circuit switched protocols. **[4]**

iii. Indicate which one would have higher latency and jitter and explain why. **[2]**

c. In contrast to TCP, UDP is described as a connectionless protocol. Briefly explain how data is transmitted between two computers using the UDP protocol. **[4]**

d. Telecommunication companies (Telcos) that provide global network services, define the services they offer to customers within a Service Level Agreement (SLA).

i. Explain the purpose of a SLA. **[1]**

ii. Give any **two (2)** examples of what it might contain. **[2]**

e. Express the following IPv6 addresses in shortened form.

i. 2001:0DB8:0000:1470:0000:0000:0000:0200 **[1]**

ii. F380:0000:0000:0000:0123:4567:89AB:CDEF **[1]**

*****END OF EXAMINATION PAPER*****

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